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Ultrasound-assisted synthesis of layered zeolitic imidazolate framework: crystal formation and characteristics

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ABSTRACT

Two-dimensional (2-D) Zn/Co, Zn, and Co zeolitic imidazolium frameworks (ZIF) were synthesized using straightforward ultrasound techniques. Dual metal Zn/Co ZIF-L, Zn ZIF-L and Co ZIF-L with leaf-like shape morphologies were prepared in 15 min. Various characterizations have been applied including X-ray diffraction, scanning electron microscopy and energy dispersive X-ray photoelectron spectroscopy to verify that the 2-D structures of these ZIF-L series have been established. The influence of various synthetic conditions such as different metal/ligand ratios as well as different synthesis times on crystal structure, morphology and porosity of Zn/Co ZIF-L were investigated. Cross-leaf morphology was obtained by reducing the metal/ligand ratio. Study of CO₂ adsorption revealed that these ZIF-L series have a considerable CO₂ adsorption capacity. In situ fabrication of TiO₂ nanoparticles within Zn/Co ZIF-L was realized via ultrasonication for 15 min as Zn/Co ZIF-L was created. TiO₂@Zn/Co ZIF-L was effectively applied as a catalyst in photodegradation of methyl orange under solar irradiation. The results indicate that ultrasound is an efficient way to synthesize high-quality metal matrix frameworks (MOFs) and immobilize nanoparticles in MOF structures with high yield, in a short time.

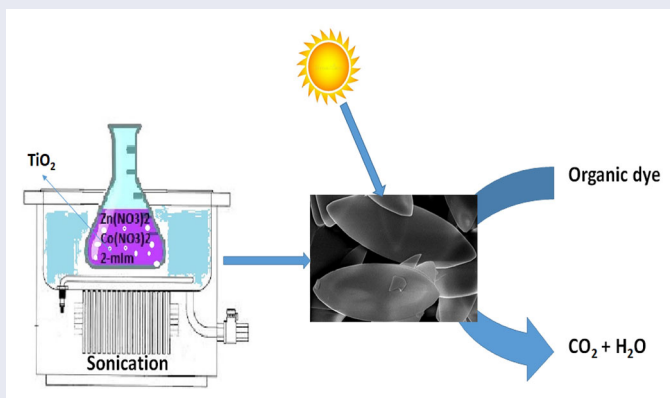
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1. Introduction

A sub-family of metal-organic frameworks (MOFs) is the zeolitic imidazolate frameworks (ZIFs), which consist of metal ions (e.g. Zn, Co, In) bridged by imidazolate ligands [1, 2]. ZIFs are synthesized as both 2-D and 3-D structures and due to their high thermal and chemical stability and porosity they have been applied in various applications including gas storage, adsorption, sensors, catalysis, etc. [3–7]. Considering a wide range of possible metal clusters and organic bridges which can be applied in the structure of MOFs, these hybrid structures provide synthetic flexibility. The crystallization process, crystal morphology, porous structure, topology, and performance of MOFs in various applications can be affected by the synthesis method [8, 9]. Generally, hydrothermal and solvothermal techniques are applied in the synthesis of various ZIFs, which mostly require long reaction times, and high temperatures and pressures [4, 10, 11]. However, from an industrial point of view, alternative methods, such as ultrasound-assisted (e.g. sonochemical) [12–15] and microwave-assisted [16] syntheses, are of interest since they enable facile, cheap, rapid, and commercially viable routes toward production of these hybrid materials. In sonochemistry, a liquid solution is irradiated by ultrasonic waves, usually generated by an ultrasonic probe immersed into the solution. As a result, acoustic cavitation may occur in the solution, which consists of the formation, growth, and collapse of bubbles, creating a temperature of $\sim 5000\text{ K}$ and a pressure of $\sim 1000\text{ bar}$, which is transient and makes a good condition for anisotropic growth of nanocrystals [17–20].

A low-cost version of ultrasound-assisted synthesis of materials is to use an ultrasonic cleaner [21] to transfer acoustic waves from the body of the cleaner to the solution. In this case, cavitation is unlikely to occur, but the ultrasonication still brings about positive effects, such as a homogeneous and faster chemical reaction. Thus, we adopted this method in this work. Various Zn-based MOFs have been synthesized by this simple approach, i.e. by using an ultrasonic bath, studying the effect of ultrasound method on size, morphology, and the growth of crystals [22, 23]. In addition, ultrasound-assisted synthesis of highly-crystalline MIL-88A particles has been carried out in an ultrasound bath so that the resultant crystals demonstrated different morphology, surface area, and porosity, arising from ultrasound conditions and reaction times [5].

ZIFs could carry or support various nanoparticles (NPs). ZIF-supported NPs could be prepared via two strategies: (1) post-synthesis of embedding NPs into as-prepared ZIFs [24, 25] and (2) in situ incorporation of NPs inside the growing ZIFs [26]. The second strategy is favorable since NPs would be surrounded by the ZIF matrix and the formation of NPs on the external surface area could be avoided, whereas undesirable deposition of NPs on the external surface in all post-synthesis approaches is unavoidable. Incorporation of NPs inside the ZIFs matrix would be an efficient way to realize separation and recovery of NPs.

In photocatalysis, photoreaction can be accelerated in the presence of catalysts. Searching for highly efficient photocatalysts is important for applications such as sunlight water splitting, disinfection of water and oxidation of organic contaminants [27]. Azo dyes, which have been widely applied in many industries, are an organic contaminant that creates environmental pollution problems. TiO_2 is a widely applied photocatalyst for degradation of several environmental contaminants including azo dyes [28–31].

Here, we report the facile, efficient, and fast ultrasound-assisted synthesis of Zn/Co, Zn, and Co ZIF-L, which are categorized as 2-D ZIF [32], and also the fabrication of TiO_2 NP@Zn/Co ZIF-L for the first time, employing an ultrasonic cleaner. This is the first report applying ultrasonic bath in the synthesis of ZIF-L in 15 min compared to 3–4 h processing time reported in the literature [32, 33]. Also, to the best of our knowledge, this is the first time that TiO_2 NPs were fabricated at ZIF-L. Fabrication of Pd NPs within the ZIF-L matrix has been reported applying Pd NPs mixed with 2-methylimidazole (2-mlm) solution in which Pd@ZIF-L was a catalyst for hydrogenation [34, 35]. In this work, we demonstrated the effectiveness of the ultrasound method in MOF synthesis as well as in situ incorporation of NPs in the MOF matrix, at the same time that MOF was produced in a very short time.

2. Experimental

2.1. Materials

Zinc nitrate hexahydrate (99%) was purchased from Sinopharm Chemical Reagent Co., Ltd. Cobalt nitrate hexahydrate (99%) was obtained from Macklin. 2-mlm (99%), methyl orange, and TiO_2 NPs (99.7%) with particle size less than 20 nm were prepared from Sigma-Aldrich. The TiO_2 NPs were dispersed in water and stirred at room temperature for 24 h, followed by drying at 100 °C to remove impurities before use. All other chemicals were used as received without purification. Deionized (DI) water was prepared in the laboratory.

2.2. Ultrasound synthesis of Zn/Co ZIF-L

To synthesize Zn, Co, and Zn/Co ZIF-L, precursors $\text{Zn}(\text{NO}_3)_2 \cdot 6(\text{H}_2\text{O})$ and/or $\text{Co}(\text{NO}_3)_2 \cdot 6(\text{H}_2\text{O})$ were dissolved in 40 mL DI water, which was then added to the 2-mlm solution under stirring. The molar ratios of Zn:Co:2-mlm are summarized in Table S1 [33]. The synthesis of ZIF-L was performed by the aid of an ultrasonic cleaner (Clangsonic, JP-030, 150 W, 40 kHz). To do so, the flask was placed in the ultrasonic bath for 15 min, 30 min, or 1 h. The resultant precipitate was collected by a centrifuge, washed with DI water several times, and finally dried in an oven at 70 °C overnight.

2.3. Room temperature (RT) synthesis of Zn/Co ZIF-L

Zn/Co ZIF-L at room temperature (RT) was synthesized by reported procedure [32, 33] to compare with the samples prepared via ultrasound in this work. $\text{M}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ ($\text{M} = \text{Zn/Co}$) and 2-mlm with molar ratios of 1:1:16, respectively, were dissolved in 40 mL DI water. After adding the metal solution to the 2-mlm aqueous solution, the mixture was stirred for 3 h. The product was collected by centrifuge, followed by washing with DI water several times and drying in an oven at 70 °C overnight. We call this reference sample as the RT sample.

2.4. Ultrasound TiO_2 NPs incorporated in Zn/Co ZIF-L (TiO_2 @ Zn/Co ZIF-L)

In a typical synthesis, 1 mL DI water was added to 0.04 mmol TiO_2 NPs and sonicated for 30 min and then the dispersion was added to the Zn/Co solution dropwise,

followed by stirring for several mins. This mixture was added to the 2-mlm solution and sonicated for 15 min in an ultrasonic bath. Finally, the resultant precipitate was collected by centrifuge, followed by drying in an oven at 70 °C overnight after washing with DI water.

2.5. Typical procedure for photocatalytic activity

A mixture of 50 mL aqueous solution of methyl orange (10 mgL^{-1}) and 20 mg of Zn/Co ZIF-L or $\text{TiO}_2@ \text{Zn/Co ZIF-L}$ was stirred for 30 min at room temperature until the adsorption–desorption equilibrium was achieved. Then the mixture was irradiated under a solar lamp (500 W) positioned 15 cm above the mixture. For the UV–vis absorption spectroscopic analysis, 3 mL aliquots were regularly withdrawn and filtered through 0.45 μm Millipore filter to remove the suspended catalyst particles, followed by determining the percentage of dye decomposition in water by monitoring the maximum absorption peak of the dye.

2.6. Characterization

X-ray diffraction (XRD) patterns of the powdered product were recorded on a Bruker D8 advanced diffractometer, applying Cu $K\alpha$ radiation within 2θ angle, ranging between 3° and 40° (3° and 70°), and a scan rate of 0.3°/s. The morphology of the resultant ZIF-L was observed by a field-emission scanning electron microscope (FE-SEM) on a Sirion 200 microscope. Energy-dispersive X-ray spectroscopy (EDS) was taken on a NOVA Nano200 SEM. Elemental analysis was performed by inductively coupled plasma-optical emission spectrometry (ICP-OES) using an iCAP7600 spectrometer. Thermogravimetric analysis (TGA) was recorded on an SDT-Q600 thermal analyzer. Sample (5 mg) was heated to 800 °C under nitrogen at a ramp rate of 10 °C/min. Specific surface areas were recorded using an Autosorb-IQ3 gas sorption analyzer under nitrogen at 77 K. CO_2 adsorption–desorption isotherm was performed on an ASAP 2020 instrument. Before gas adsorption analysis, all samples were pretreated at 383 K under vacuum for 3 h. X-ray photoelectron spectroscopy (XPS) was recorded on an ESCALAB 250 X-ray photoelectron spectrometer (Thermo Fisher Scientific, Waltham, MA, USA) equipped with monochromated Al $K\alpha$ radiation ($h\nu = 1486.6 \text{ eV}$). The C1s peak (284.8 eV) was used for the calibration of binding energy. Transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) were taken by a field-emission transmission electron microscope, TALOS F200X. Since the sample was magnetic, it was cut into thin slices by applying an ultramicrotome (Leica EM UC7) before TEM analysis. UV–vis absorption spectra of methyl orange aqueous solutions were recorded on a UV/VIS/NIR spectrophotometer (Lamda 950).

3. Discussion

3.1. Crystal structure, morphology, and structural properties of Zn/Co ZIF-L

By using an ultrasonic bath, Zn/Co ZIF-L was synthesized in times of 15 min, 30 min, and 1 h. The morphology of the product was studied by SEM. Figure 1(a–d) displays

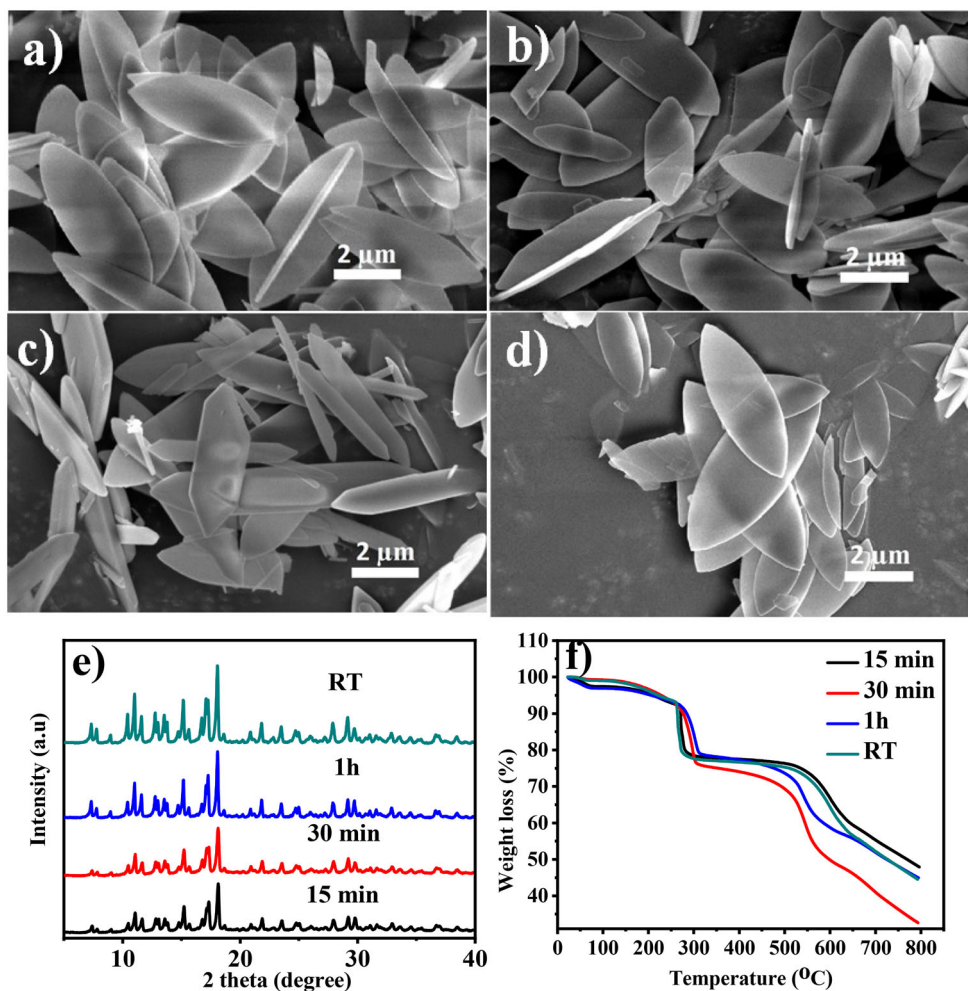


Figure 1. SEM images of Zn/Co ZIF-L synthesized by ultrasound technique at various times: (a) 15 min, (b) 30 min, (c) 1 h, and (d) at RT in 3 h (reference sample). (e) XRD patterns and (f) TG curves of Zn/Co ZIF-L, synthesized by ultrasound technique at 15 min, 30 min, 1 h and at RT in 3 h (reference sample).

the leaf-like shape of the Zn/Co ZIF-L, which was formed by the ultrasound technique at different times, as well as the reference or RT sample. Zn/Co ZIF-L structure was formed within 15 min via ultrasound (as confirmed by XRD as well, Figure 1(e)). However, it seems that by increasing the ultrasonication time from 15 min to 1 h, the morphology is changed from leaf-like (Figure 1(a)) to hexagonal (Figure 1(b,c)). Moreover, the crystals which were formed in 30 min and 1 h look slender, compared to those formed in 15 min, and there is more variety in the size of crystals in these two samples. Both Zn/Co ZIF-L formed in 15 min under ultrasound and the one generated at RT exhibit leaf-like shape morphology. This leaf-like shape morphology is in accord with the one reported for ZIF-L [32, 33].

The powder XRD patterns confirmed the phase purity of the resultant crystals. The XRD spectra for all samples are identical and are in excellent agreement with the

Table 1. Zn and Co mass ratio in the Zn/Co ZIF-L structure and the yield based on Zn for various samples.

Process condition	Zn ratio	Co ratio	Yield (%)
Ultrasound-15 min	53	47	90
Ultrasound-30 min	54	46	87.6
Ultrasound-1 h	54	46	77
RT-3 h	53	47	74

literature (Figure 1(e)) [32, 33]. These powder XRD patterns reveal synthesis of ultrasound-assisted Co/Zn ZIF-L structure at short times. Highly crystalline patterns were acquired for all samples and clearly, no impurities were observed, confirming the effectiveness of the ultrasound technique to synthesize crystalline structures with high purity. It is clear that the ultrasound effect provides enough energy to form a crystalline framework.

As displayed in the TG curves of Figure 1(f), for all ultrasonicated samples, and the reference RT sample, the weight is reduced slowly below 260 °C, followed by a sharp weight loss at 266 °C, which is supposedly due to the loss of guest water molecules and feebly linked 2-mlm [32]. At 266 °C, the mass residues for Zn/Co ZIF-L prepared within 15 min, 30 min, and 1 h under ultrasonication and RT are 77.7, 74.7, 77.5, and 76.8%, respectively. There is a long plateau at 280–520 °C for samples prepared in 15 min of ultrasonication and at RT, and at 300–500 °C for samples prepared in 30 min and 1 h of ultrasonication, revealing the high thermal stability of these samples in the absence of guest molecules. Zn/Co ZIF-L synthesized within 15 min of ultrasonication possesses the same thermal stability as Zn/Co ZIF-L synthesized at RT in 3 h, consistent with the fact that they both have the same SEM morphology. Thus, the ultrasound technique significantly decreases the processing time of the ZIF-L product, while its characteristics are preserved.

We investigated the presence and distribution of Zn and Co in the samples by ICP and EDS mapping. According to the ICP results listed in Table 1, the Zn:Co mass ratio is almost 50:50, the same as the ratio which was applied for their synthesis (Table S1), although the contribution of Zn is slightly higher than Co. Obviously, the Zn/Co ZIF-L samples synthesized by applying the ultrasound technique had a higher yield (based on Zn), corroborating the efficiency of this technique for synthesizing MOFs (Table 1). In addition, SEM-EDS images (Figures S1–S3, Supplementary Information) indicate the homogeneous dispersion of metals in the framework of Zn/Co ZIF-L synthesized by ultrasound at various times. Noteworthy, the two metals were randomly distributed in the crystal structure.

Porosity properties of Zn/Co ZIF-L synthesized via ultrasound technique and at RT were analyzed by nitrogen physisorption at 77 K; Figure 2(a) displays the nitrogen physisorption isotherms. The isotherm of all Zn/Co ZIF-L samples, either synthesized via ultrasound or stirred at RT appeared as IUPAC type II isotherm [36], suggest this ZIF-L is a nonporous material. The Langmuir, BET, and micropore volume of these Zn/Co ZIF-L samples derived from the related isotherm are summarized in Table S2. It seems that applying different synthesis methods (ultrasound or RT) does not affect the surface area and porosity properties of Zn/Co ZIF-L samples.

Figure 2(b) exhibits CO₂ adsorption–desorption isotherms of Zn/Co ZIF-L samples, performed at 298 K. All samples, i.e. synthesized via ultrasound and at room

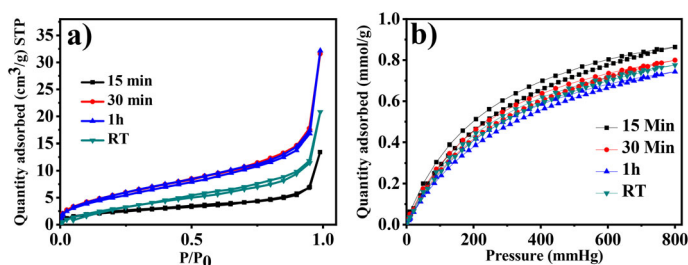


Figure 2. (a) N_2 adsorption isotherms at 77 K and (b) CO_2 adsorption-desorption isotherms at 273 K of Zn/Co ZIF-L, synthesized by ultrasound at 15 min, 30 min, and 1 h and at RT in 3 h (reference sample).

temperature, exhibit similar CO_2 adsorption capacity. However, CO_2 uptake by Zn/Co ZIF-L synthesized via ultrasound within 15 min (Figure 2(b), 0.87 mmol g^{-1}) is slightly higher compared to those processed for 30 min (Figure 2(b), 0.80 mmol g^{-1}), 1 h (Figure 2(b), 0.75 mmol g^{-1}) and even RT (Figure 2(b), 0.78 mmol g^{-1}). In spite of the fact that the Zn/Co ZIF-L has much lower porosity and surface area in comparison to other ZIFs (Table S3, SI), the CO_2 uptake capacity of this 2-D Zn/Co ZIF-L exceeds that of ZIF-8, -71, and -95, and is comparable to that of ZIF-100, -25, -97, and MOF-5 (Table S3, SI). ZIF-L possesses a 2-D layered structure, which is part of sodalite (SOD) topology found in 3D ZIF-8. The layers in ZIF-L are parallel to each other, stacking by the hydrogen bonds between 2-mlm molecules. Since ZIF-L does not have a channel system like ZIF-8, it is believed that CO_2 adsorption by ZIF-L is associated with gate opening effect within the structure, as proposed for ZIF materials [37]. The interesting point about MOFs compared to other porous materials is that the major portion of the MOF building unit is composed of organic moieties, which generate highly flexible frameworks. Gas diffusion or adsorption of Zn/Co ZIF-L is benefited from flexible frameworks. Figure S4 displays three cycles of CO_2 adsorption for Zn/Co ZIF-L, synthesized by ultrasonication within 15 min, which indicates the reversibility of the CO_2 adsorption. It is clear that the second and third cycles share the same amount of CO_2 adsorption as the first one, and there is no deterioration of adsorption capacity. As shown MOF nano-sheets are good examples for gas separation [3], we believe that this 2-D Zn/Co ZIF-L structure is an ideal choice to prepare membranes for gas sensing or gas separation.

3.2. Synthesis conditions of Zn/Co ZIF-L

Generally, the properties of ZIF materials including morphology, surface area, porosity, and particle size could be influenced by the synthesis conditions. We investigated various parameters such as the ratio of metal salt to the ligand and the time of synthesis via ultrasound technique. Increasing the number of nuclei during MOF synthesis can accelerate crystallization, resulting in a shorter time in crystal growth [38]. The crystallization rate in Zn/Co ZIF-L is increased via employing the ultrasound technique as the crystallization time is decreased. Therefore, to study the crystallization rate of Zn/Co ZIF-L, the concentration of the starting solution was changed by altering the ratio of metal salt (Zn/Co) to ligand (2-mlm). Accordingly, the amount of solvent and metal

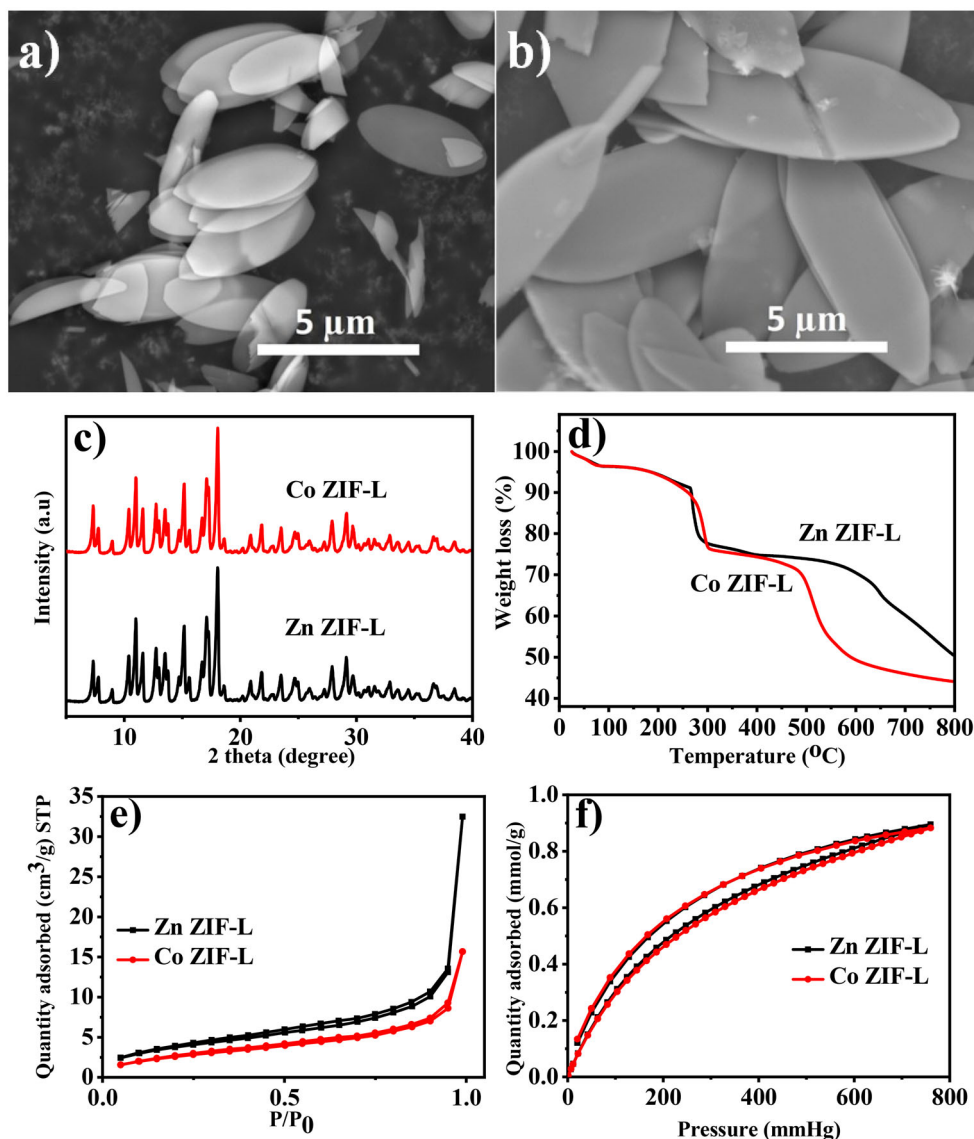


Figure 3. The SEM images of (a) Zn and (b) Co ZIF-L synthesized by ultrasound technique in 15 min, (c) XRD patterns, (d) TG curves, (e) N_2 adsorption isotherms at 77 K, and (f) CO_2 adsorption-desorption isotherms at 273 K of Zn and Co ZIF-L, synthesized by ultrasound technique in 15 min.

precursors were kept constant, while the amount of 2-mlm was changed. In this investigation, we studied the morphology, crystallinity, and surface areas of the three samples of Zn/Co ZIF-L synthesized via ultrasound with various metal/ligand ratios and also in various synthesis times (Table S4). Decreasing the metal/ligand ratio resulted in the formation of hierarchical ZIF-L structures (Figure S5(a-c) of SI). This cross leaves-shaped morphology is different from leaf-like morphology which was obtained for the sample with metal/ligand ratio 2:16 (Figure 1(a)). As Figure S5(a-c) exhibits, the number of cross leaves-shaped ZIF-L in the sample with metal/ligand ratio 2:8 (sample 2) is

more than in the one with ratio 2:12 (sample 1). Also, SEM image of the sample with metal/ligand ratio 2:8 synthesized via ultrasound within 1 h (sample 3, [Figure S5\(c\)](#)) shows some circular flakes besides the cross leaves-shaped morphology, and clearly, the size of leaves in this sample is smaller than those in samples 1 and 2. XRD patterns of samples 1–3 ([Figure S5\(d\)](#)) are similar and fitted well with the one with ratio 2:16 ([Figure 1\(e\)](#)), confirming the formation of Zn/Co ZIF-L. However, the peak at $2\theta = 11.5^\circ$ for sample 3 (metal/ligand ratio 2:8 ultrasonicated for 1 h) is slightly broadened which is probably due to formation of the circular flakes observed in the SEM image ([Figure S5\(c\)](#)). N_2 isotherms for samples 1–3 are displayed in [Figure S5\(e\)](#), suggesting that these 3 samples are non-porous. Table S5 shows the Langmuir, BET, and micropore volume of samples 1–3. Interestingly, sample 3 (with metal/ligand ratio 2:8 synthesized via ultrasound in 1 h) demonstrated higher surface area and bigger pore volume. Also, as the ratio decreased, the yield of product decreased (79%, 52.6%, and 49.7% are the yields of samples 1, 2, and 3, respectively). Consequently, morphology and particle size are extremely dependent on the molar ratio of metal/ligand. As metal/ligand ratio decreases cross leaves morphology is observed. Therefore, more 2-mlm is favorable for formation of the product with a single leaf-like shape morphology with high yield.

3.3. Crystal structure, morphology, and structural properties of Zn and Co ZIF-L

With the successful synthesis of ZIF-L with metal/ligand ratio 2:16 showing leaf-like morphology processed for 15 min via ultrasound technique, Zn ZIF-L and Co ZIF-L were also prepared via this method at identical synthesis conditions. As illustrated in [Figure 3\(a,b\)](#), both Zn ZIF-L and Co ZIF-L show leaf-like shape morphology, similar to that of Zn/Co ZIF-L. Noteworthy, comparison of the crystal size of these Zn, Co, and Zn/Co ZIF-L microstructures reveal that the Zn ZIF-L consists of smaller crystals, while Co ZIF-L shows larger crystal size. XRD patterns ([Figure 3\(c\)](#)) demonstrate the high crystallinity of these ZIF-Ls, indicating all peaks match well with Zn/Co ZIF-L ([Figure 1\(e\)](#)) as well as the literature [32]. Because the synthesis conditions for Zn/Co ZIF-L, Zn ZIF-L and Co ZIF-L were the same and just the metal salt was different, and given that their crystallinity is similar, it is inferred that these ZIF-L materials are isostructural. Also, EDS mapping indicates good distribution of Zn and Co in Zn ZIF-L and Co ZIF-L, respectively, as illustrated in Figures S6 and S7. These ZIF-Ls were obtained in high yields (87.7% and 85% for Zn ZIF-L and Co ZIF-L, respectively), confirming the high throughput of the ultrasound technique for the synthesis of these ZIF-Ls. [Figure 3\(d\)](#) displays that the TG curves of Zn ZIF-L and Co ZIF-L are similar and agree with the TG curve of Co/Zn ZIF-L, synthesized within 15 min. For both Zn ZIF-L and Co ZIF-L structures, there is a gradual weight loss to 260 °C and sharp weight loss up to 290 °C, corresponding to loss of guest water molecules and the weakly linked 2-mlm, followed by a long plateau up to 580 °C for Zn ZIF-L and 500 °C for Co ZIF-L, indicating high thermal stability of these structures. Zn ZIF-L exhibited higher thermal stability than Co ZIF-L. N_2 isotherms for Zn and Co ZIF-L demonstrate that these samples are not porous, either ([Figure 3\(e\)](#)). Table S6 shows the Langmuir, BET, and micropore volume of these Zn and Co ZIF-L structures. There is no big difference in surface areas and

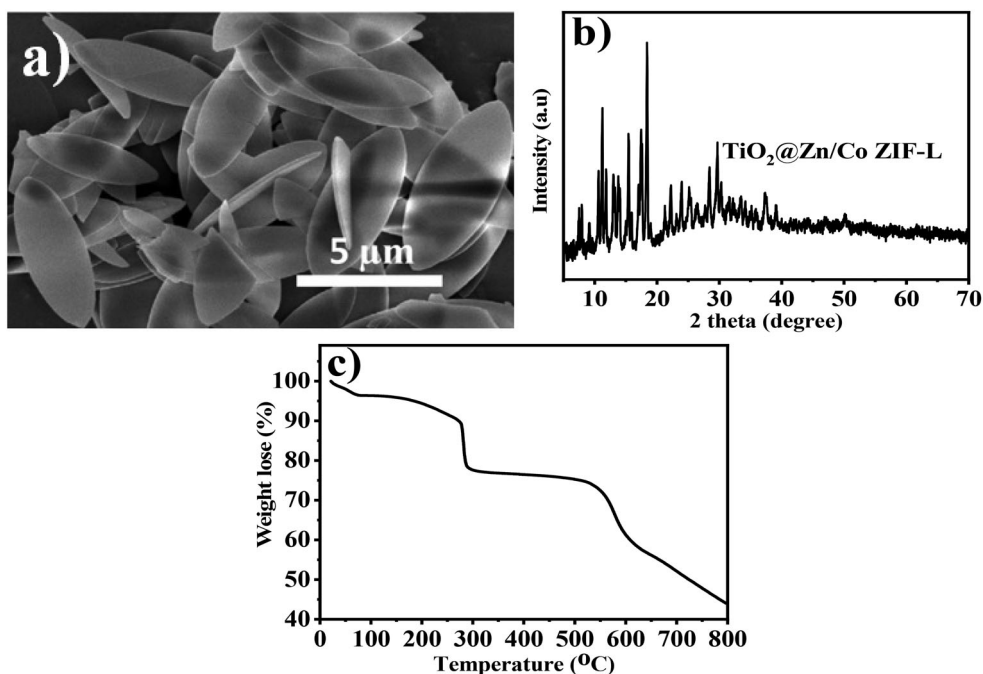


Figure 4. (a) The SEM images, (b) XRD, and (c) TG curve of $\text{TiO}_2@\text{Zn/Co ZIF-L}$ synthesized by ultrasound technique within 15 min.

porosity properties of Zn, Co, and Zn/Co ZIF-L. CO_2 adsorption–desorption was performed on Zn and Co ZIF-L at 298 K, as illustrated in Figure 3(f). Both Zn and Co ZIF-L shared similar CO_2 capacity (0.89 mmol g^{-1}), which was also comparable to the CO_2 uptake of Zn/Co ZIF-L.

3.4. Physical-chemical properties of $\text{TiO}_2@\text{Zn/Co ZIF-L}$

To study the potential of the ultrasound technique for embedding of NPs within ZIF-L, we chose TiO_2 NPs. The morphology and crystallinity of the TiO_2 NPs were first analyzed. As depicted in Figure S8(a), the size of TiO_2 NPs is less than 20 nm and XRD patterns exhibit all peaks related to the anatase crystalline phase [39], as shown in Figure S8(b). These TiO_2 NPs were added to the metal solution of ZIF-L, which was then added to the ligand solution followed by ultrasonication for 15 min. Figure 4(a) reveals the leaf-like shape morphology of $\text{TiO}_2@\text{Zn/Co ZIF-L}$, which confirms preserving the morphology of Zn/Co ZIF-L after TiO_2 loading. Moreover, the XRD pattern of $\text{TiO}_2@\text{Zn/Co ZIF-L}$ (Figure 4(b)) is in excellent agreement with Zn/Co ZIF-L, indicating that the crystalline structure of Zn/Co ZIF-L has been maintained after TiO_2 loading. Nevertheless, no obvious peaks of TiO_2 ($2\theta = 25^\circ$) were found in the XRD, probably due to good dispersion of NPs and/or low concentration of TiO_2 . There is a possibility that the TiO_2 peak at $2\theta = 25^\circ$ (101) overlapped with Zn/Co ZIF-L peak in this region. No other attributable peaks of impurities or any other phase can be recognized, suggesting high crystallinity of the product. Additionally, the TG curve of $\text{TiO}_2@\text{Zn/Co ZIF-L}$ is shown in Figure 4(c), exhibiting the same trend in weight loss as obtained for Zn/

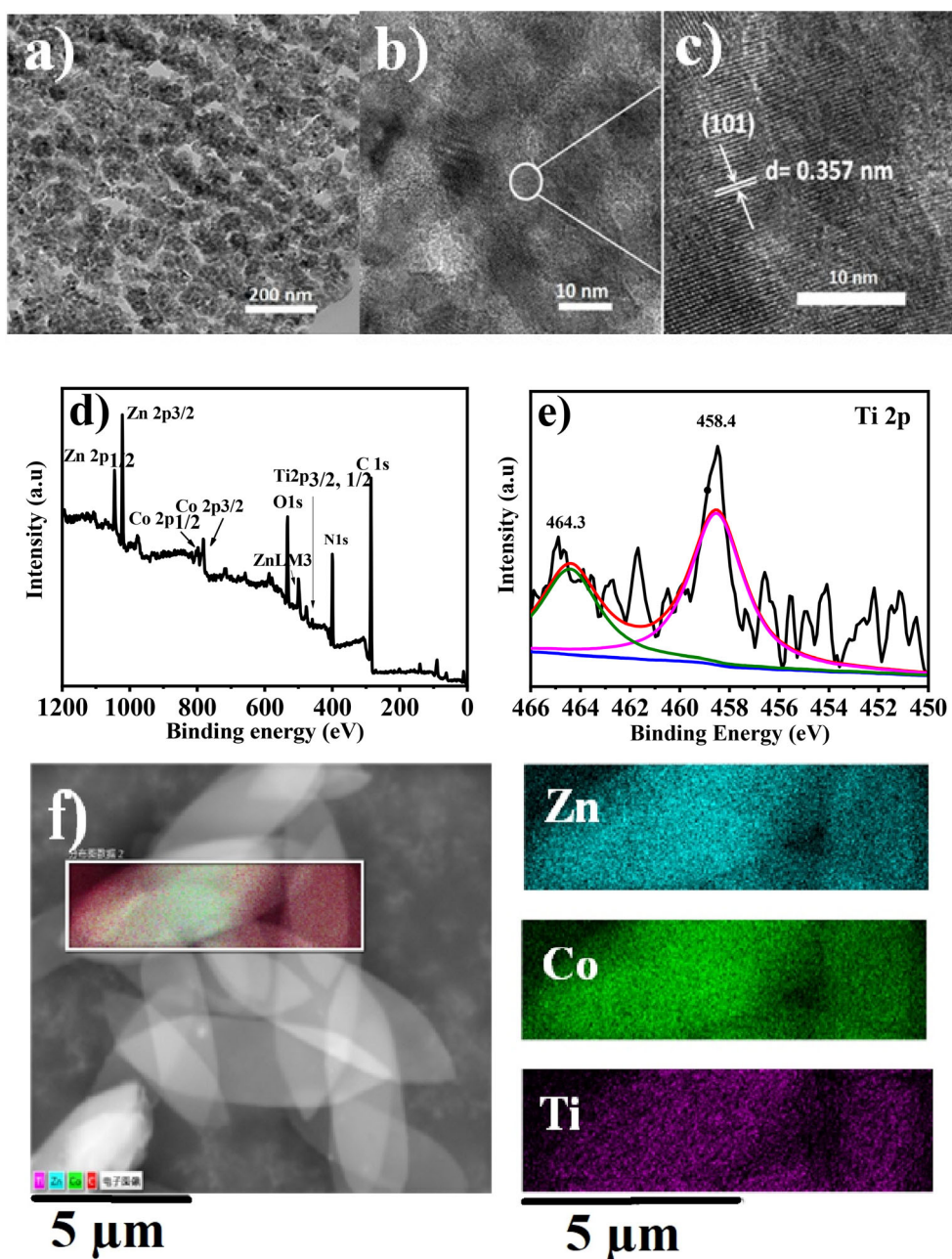


Figure 5. (a) TEM, (b) HRTEM of $\text{TiO}_2@\text{Zn/Co}$ ZIF-L, (c) lattice fringes of TiO_2 incorporated Zn/Co ZIF-L indicating anatase crystalline phase of TiO_2 , (d) general survey XPS analysis, (e) binding energies of Ti 2p. (f) SEM image of $\text{TiO}_2@\text{Zn/Co}$ ZIF-L (left) and distribution of Zn, Co, and Ti in this sample (right).

Co ZIF-L (a sharp weight loss at 277°C due to losing water molecules inside the structure and weakly connected 2-mlm followed by a long plateau up to 530°C), indicating high thermal stability of this structure, as well.

TEM and HRTEM were performed to analyze the size and morphology of TiO_2 NPs @Zn/Co ZIF-L (Figure 5(a–c)). As presented in Figure 5(a), no aggregation of TiO_2 NPs was found in the MOF structure, indicating good dispersion of TiO_2 NPs in the ZIF-L structure, which strongly confirms the efficiency of the ultrasound technique in the assembly of NPs in ZIF-L structure. HRTEM investigation of TiO_2 @Zn/Co ZIF-L structure (Figure 5(b,c)) presents a high crystallinity of TiO_2 NPs and also shows lattice spacing that we can assign to the distance between the adjacent lattice fringes. The lattice spacing is $d = 0.357$ nm, which is consistent with the lattice spacing of the (101) plane of the anatase crystalline phase of TiO_2 [40, 41]. Furthermore, HRTEM (Figure 5(b)) shows that the size of the TiO_2 NPs is less than 20 nm as observed in the SEM image (Figure S8(a)). The quantitative amount of TiO_2 in TiO_2 @Zn/Co ZIF-L was determined as 0.23%wt by ICP analysis. Furthermore, the surface of TiO_2 @Zn/Co ZIF-L was analyzed by X-ray photoelectron spectroscopy (XPS). A general survey scan of XPS analysis, presented in Figure 5(d), reveals the presence of Co, Zn, C, N, O, and Ti in Zn/Co ZIF-L incorporating TiO_2 . Also, the high-resolution spectra of TiO_2 @Zn/Co ZIF-L (Figure 5(e)) demonstrate two peaks at 458.68 eV and 464.8 eV, corresponding to the $\text{Ti}2p_{3/2}$ and $\text{Ti}2p_{1/2}$, respectively, which is in excellent agreement with values previously reported for Ti with +4 oxidation state [27, 42]. The binding energies for Zn $2p_{3/2}$ and $2p_{1/2}$ peaks appeared at 1020.40 and 1044.98 eV, respectively, and Co $2p_{3/2}$ and $2p_{1/2}$ peaks appeared at 782.10 and 797.65 eV, respectively (Figure 5(d)), which agree well with literature data for Zn(II) and Co(II) [6, 43]. To further confirm the incorporation of TiO_2 nanoparticles within the Zn/Co ZIF-L, a mapping of TiO_2 @Zn/Co ZIF-L was executed employing EDS. Figure 5(f) demonstrates the homogeneous dispersion of Ti on the surface of ZIF-L.

The above results indicate TiO_2 NPs were embedded in Zn/Co ZIF-L structure, indicating the effectiveness of the ultrasound technique in assembling of NPs in ZIF-L structure in one-step and short processing time of 15 min.

3.5. Photocatalytic activity of TiO_2 @Zn/Co ZIF-L

The degradation of Methyl orange (MO) in the presence of prepared ZIF-L samples under solar light has been studied to evaluate the photocatalytic performance of these prepared ZIF-L samples. Methyl orange (MO) is an organic azo dye (the structure is displayed in Figure S9(d)), widely used in the industrial colorant. Photocatalysis of organic dyes like MO to purify wastewater has received much attention. Several researchers reported the degradation of MO by applying TiO_2 as a catalyst, including Heon Lee *et al.* [44], who applied the $\text{TiO}_2/\text{K}_2\text{S}_2\text{O}_8$ hybrid system irradiated under microwave and UV. Also, Shang *et al.* [45] showed the effectiveness of commercial TiO_2 in the degradation of MO under visible light. In this study, photocatalytic properties of TiO_2 @Zn/Co ZIF-L catalyst were investigated by a model reaction of degradation of MO as a pollutant in an aqueous solution under solar irradiation. Zn/Co ZIF-L was also tested under the same experimental condition, as the reference material to characterize the photocatalytic activity of Zn/Co ZIF-L and TiO_2 @Zn/Co ZIF-L. The effect of TiO_2 nanoparticles which are immobilized in the Zn/Co ZIF-L structure in the photodegradation of MO under solar irradiation would be evident. MO exhibits a strong absorption at 664 nm,

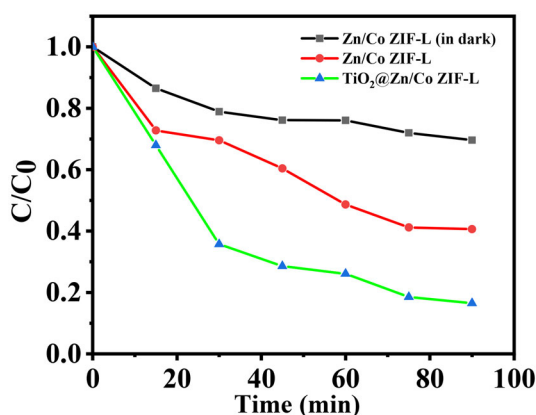


Figure 6. Decay of MO concentration in the presence of Zn/Co ZIF-L (without light), and Zn/Co ZIF-L and TiO₂@Zn/Co ZIF-L as catalysts under solar irradiation.

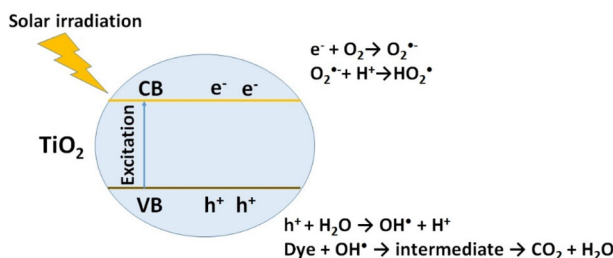


Figure 7. The schematic possible mechanism for dye degradation by TiO₂ under solar irradiation.

and a change in its maximum absorbance is a measure of dye degradation (Figure S9). Figure 6 demonstrates the significant decay in MO concentration was obtained when Zn/Co ZIF-L and TiO₂@Zn/Co ZIF-L were applied. As C and C₀ represent the concentration of MO at any given time and initial concentration, respectively. While the degradation of MO reached 30% in 90 min in the presence of Zn/Co ZIF-L in the dark, Zn/Co ZIF-L and TiO₂@Zn/Co ZIF-L achieved 59% and 84% degradation of MO under solar irradiation in 90 min, respectively. The incorporation of TiO₂ at Zn/Co ZIF-L led to an increase in the photocatalytic activity of Zn/Co ZIF-L.

The possible mechanism of MO degradation by TiO₂ NPs could be explained by photogeneration of electron-hole pairs between the conduction band and valence band due to excitation of TiO₂ NPs under solar light (Figure 7). By creating e[−]–h⁺ pair separation, the generation of active radicals for an organic pollutant degradation could occur by the mechanism shown in Figure 7. In this mechanism OH[•] attacks organic compounds, resulting in the formation of reaction intermediates, which further react with OH[•] to produce final degradation products, such as CO₂ and H₂O.

Since the stability of the heterogeneous catalyst is an issue, the structural morphology and stability of recovered TiO₂@Zn/Co ZIF-L were analyzed. The SEM image of the recovered TiO₂@Zn/Co ZIF-L (Figure S10(a)) shows that the morphology of the catalyst has been preserved after the reaction. The XRD pattern of Figure S10(b) shows all distinctive diffraction peaks, confirming that the crystallinity of the catalyst has been preserved after reaction and no significant crystallographic degradation has

occurred. Moreover, according to the ICP-AES analysis, no trace of Ti was found in the filtered reaction mixture, suggesting that the TiO_2 NPs could be retained within the ZIF-L structure during the reaction. These results indicate that $\text{TiO}_2@\text{Zn/Co}$ ZIF-L as photocatalyst is stable and can be applied multiple times.

4. Conclusion

We have presented a facile, rapid, and efficient synthesis of Zn/Co, Zn, and Co ZIF-L, sub-class of metal-organic frameworks (MOFs), with the aid of ultrasound technology in a simple ultrasonic bath in 15 min with high yields. These Zn/Co, Zn, and Co ZIF-L materials showed high CO_2 adsorption capacity. Zn/Co ZIF-L was synthesized in various ratios of metal/ligand and various synthesis times applying the ultrasound technique to investigate the effect of synthetic reaction conditions on the morphology and crystalline structure of Zn/Co ZIF-L. Interestingly, by decreasing the metal/ligand ratio, the crystalline structure of ZIF-L was preserved, while cross-leaves type morphology was obtained. TiO_2 nanoparticles were fabricated within the Zn/Co ZIF-L via ultra-sonication in one-step as ZIF-L was formed. The $\text{TiO}_2@\text{Zn/Co}$ ZIF-L was efficiently employed as a photocatalyst in the degradation of MO dye under solar irradiation. This fast and environmentally friendly ultrasound route not only can manufacture MOF materials on a large scale but can also be used to introduce nanoparticles within the MOF structure at the same time that MOF was formed. Here, we applied TiO_2 NPs as a type of NPs that can be fabricated through MOF structure. However, as concluded from this study, other NPs such as metal NPs also could be immobilized within the MOF structure.

Disclosure statement

No potential conflict of interest was reported by the authors.

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